

12.07 Random Variables and Data

1 For each random variable, state whether it is discrete or continuous.

- a** A film critic's rating of a film, from 0 to 4 stars.
- b** The speed of a car.
- c** The sum rolled on a pair of dice.
- d** The winning ticket number drawn from a raffle.
- e** The weight of parcels at a post office.
- f** The size of jeans in a shop.
- g** The temperature of a metal as it cools.
- h** The amount of water in different types of fruit drink.
- i** The number of cars passing the school over a 10-minute period.
- j** The number of cities in each country in Europe.
- k** The number of heads when tossing a coin 50 times.
- l** The number of correct answers in a 10-question test.

2 Write the set of possible values for each discrete random variable.

- a** Number of daughters in a one-child family.
- b** Number of 6s on 10 rolls of a die.
- c** Number of people aged over 50 in a group of 20 people.
- d** The number of days it rains in March.
- e** The sum of the 2 numbers rolled on a pair of dice.

3 For each set of data below:

- i** draw a frequency distribution table
- ii** draw a histogram and frequency polygon
- iii** find the highest and lowest scores (groups for part **d**)
- iv** find the mode (group for part **d**).

a Results of a class quiz:

8, 6, 5, 7, 6, 8, 3, 2, 6, 5, 8, 4, 7, 3, 8, 7, 5, 6, 5, 8, 6, 4, 9, 6, 5

b The number of people ordering pizzas each night:

15, 12, 17, 18, 18, 15, 16, 13, 15, 17, 18, 12, 17, 14, 16, 15, 17, 18, 19, 15, 15, 12

c The number of people attending a gym:

110, 112, 114, 109, 112, 113, 108, 110, 113, 112, 113, 110, 109, 110, 110, 112, 114, 114, 112, 114, 113

d The heights of students (in cm) in a Year 12 class:

159, 173, 182, 166, 172, 179, 181, 163, 178, 169, 183, 158, 162, 167, 174, 175, 180, 174, 176, 159, 161, 171, 174, 179, 180, 159, 157

(Use groups of 155–159, 160–164, 165–169 and so on.)

- 4 For each data set:
- i add a cumulative frequency column and class centre where necessary
 - ii sketch a cumulative frequency histogram and ogive (cumulative frequency polygon).

a Number of cars in a school car park:

Number of cars	Frequency
10	4
11	8
12	11
13	9
14	5

b Results of a science experiment:

Score	Frequency
1	7
2	1
3	3
4	0
5	2
6	5

- 5 The table shows the number of daily rescues at a beach over a period of time.

- a Draw a histogram showing this data.
- b How many times were more than 6 rescues made?
- c What was the mode number of daily rescues during the survey?

Number of rescues	Frequency
4	1
5	3
6	6
7	4
8	5
9	0
10	2

- 6 For each data set, find:

- i the mode
- ii the median.

- a Number of people auditioning for parts in a play: 5, 5, 7, 6, 5, 6, 1
- b Number of minutes for an ambulance to respond to a call: 1, 4, 6, 8, 7, 4, 6, 4, 5
- c Ages of students on a basketball team: 15, 18, 14, 19, 18, 17, 11
- d Scores on a class quiz: 4, 6, 5, 4, 7, 8

- 7 For each data set, find:

- i the median
- ii the mode.

a Judge's scores on a dance contest:

Score	Frequency
3	3
4	4
5	2
6	7
7	6
8	2
9	3

b Number of matches in each match box surveyed:

Number of matches	Frequency
50	1
51	6
52	5
53	3
54	4
55	2

c Mark in a History assignment:

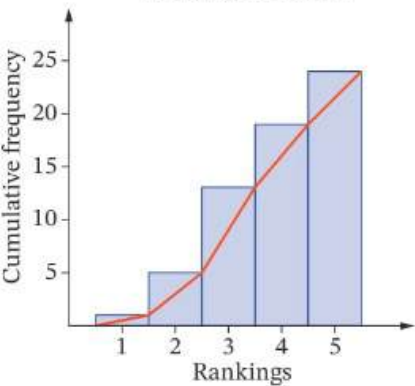
Mark	Frequency
14	4
15	2
16	1
17	4
18	3
19	5
20	6

d Attendances at hockey matches:

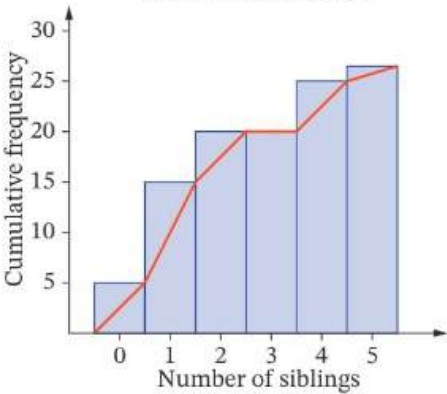
Attendance	Frequency
100	3
101	0
102	2
103	1
104	6
105	5

8 Find the median from each ogive:

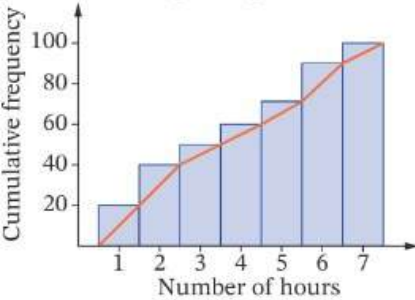
a Ranking of a film



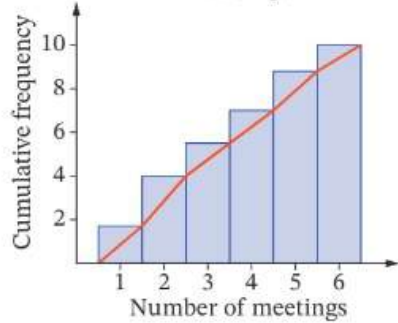
b Number of siblings



c Hours of guitar practice a week



d Attendance at social committee meetings



9 Find the mode of each data set represented by the ogives in Question 8.

10 For each data set:

- i add a cumulative frequency column
- ii draw a cumulative frequency polygon
- iii find the median from the graph (estimate for parts c and d).

a Number of athletes representing their school over a 30-year period:

Number of athletes	Frequency
1	5
2	6
3	4
4	8
5	5
6	2

b Number of lollies in a bag:

Number of lollies	Frequency
45	3
46	5
47	1
48	7
49	3
50	1

c Hours a week worked by employees in a cafe:

Number of hours	Frequency
1–5	7
6–10	5
11–15	3
16–20	6
21–25	7
26–30	2

d Time to complete a race:

Time (min)	Frequency
2.5–2.8	3
2.9–3.2	2
3.3–3.6	0
3.7–4.0	6
4.1–4.4	1
4.5–4.8	4
4.9–5.2	4

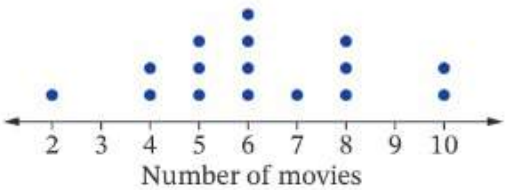
11 For each data set:

- i draw a frequency distribution table including cumulative frequency
- ii find the mode
- iii draw an ogive
- iv find the median from the ogive

a Home runs scored over a baseball season:

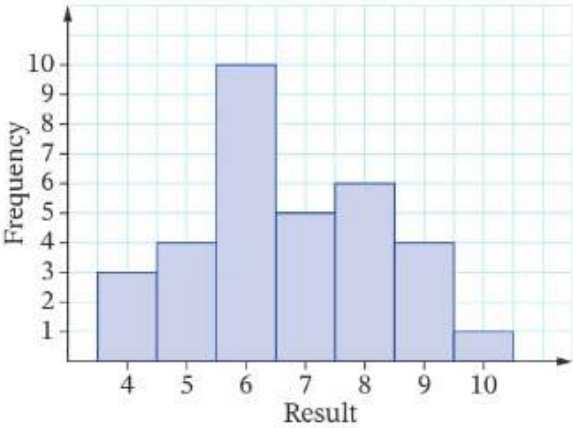
4, 6, 5, 8, 8, 6, 5, 3, 4, 9, 6, 3, 5, 6, 5, 4, 7, 5, 8, 5, 6, 2, 3

b Number of movies seen in a year:



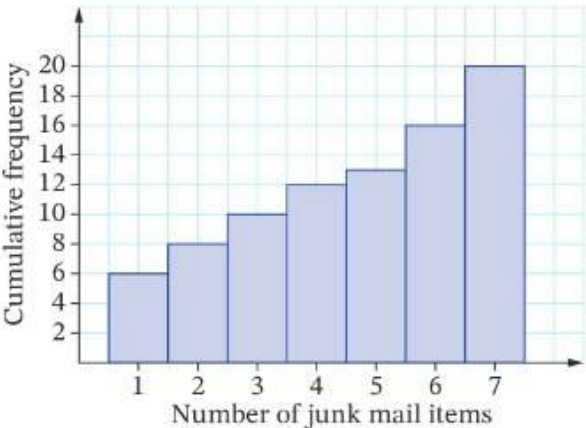
12 The frequency histogram shows the results on a maths quiz.

- a** Find the mode.
- b** Draw a cumulative frequency histogram and polygon and use them to find the median.



13 The cumulative frequency histogram shows data collected in a survey on the number of junk mail items that people received daily in their inbox.

- a** Draw a frequency distribution table to show the number of junk mail items people receive daily.
- b** Construct a frequency histogram to show this data.
- c** Find the mode and the median of this data set.



12.08 Relative Frequency and Probability

1 This table shows the scores that a class earned on a music performance.

- a** Find the relative frequency for each score in the table, in fraction form.
- b** If a student is chosen at random from this class, find the probability that this student:
 - i** scored 8
 - ii** scored less than 7
 - iii** scored 5 or more.
- c** What score is:
 - i** most likely?
 - ii** least likely?

Score	Frequency
4	6
5	4
6	1
7	7
8	2
9	3

2 The table shows the results of a survey into the number of days students study each week.

- a** Find the relative frequency as a percentage for each number of days.
- b** If a student was selected at random, find the most likely number of days this student studies.
- c** Find the probability that this student would study for:
 - i** 1 day
 - ii** 5 days
 - iii** 3 or 4 days
 - iv** at least 4 days
 - v** fewer than 3 days.

Number of days	Frequency
1	3
2	6
3	1
4	7
5	2
6	1

3 The table shows the results of a trial HSC exam.

- a** Calculate the relative frequency as a decimal for each class.
- b** Find the probability that a student chosen at random from these students scored:
 - i** between 20 and 39
 - ii** between 60 and 99
 - iii** less than 40.

Class	Frequency
0–19	9
20–39	12
40–59	18
60–79	7
80–99	4

4 This table shows the results of a science experiment to find the velocity of an object when it is rolled down a ramp.

- a** Write the relative frequency of each velocity as a fraction.
- b** Find the probability that an object selected at random rolls down the ramp with a velocity between:
 - i** 5 and 7 m/s
 - ii** 11 and 13 m/s
 - iv** 11 and 16 m/s
 - v** 2 and 10 m/s.
- c** Find the probability that the object has a velocity:
 - i** less than 8 m/s
 - ii** 5 m/s or more

Velocity (m/s)	Frequency
2–4	2
5–7	7
8–10	4
11–13	1
14–16	6

iii 8 and 10 m/s

iii more than 7 m/s.

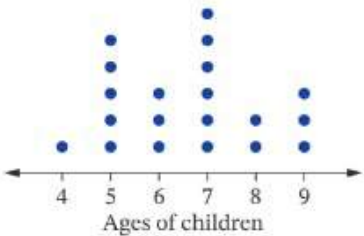
- 5** A telemarketing company records the number of sales it makes per minute over a half-hour period. The results are in the table.
- a** What percentage of the time were there 3 sales per minute?
 - b** Write the relative frequencies as percentages.
 - c** What is the most likely number of sales/minute?
 - d** Find the probability of making:
 - i** 2 sales/minute
 - ii** 5 sales/minute
 - iii** more than 2 sales per minute.

Sales/min	Frequency
0	4
1	12
2	6
3	3
4	0
5	5

- 6** **a** The waiting times in minutes between trains arriving at a station are shown below. Organise the scores below in a frequency distribution table, including relative frequency and cumulative relative frequency columns.

9, 5, 4, 7, 7, 9, 4, 6, 5, 8, 9, 6, 7, 4, 4, 3, 8, 5, 6, 9

- b** Draw a relative frequency histogram for this data.
 - c** Find the probability that the waiting time between trains is:
 - i** 7 minutes
 - ii** at least 8 minutes
 - iii** less than 5 minutes
 - iv** 7 minutes or less.
- 7** **a** From the dot plot of ages of children at a fun park, draw up a frequency distribution table.
- b** Draw a relative frequency polygon for this data.
 - c** Find the probability (as a decimal) that a child chosen at random from the fun park is aged:
 - i** 8
 - ii** at least 6
 - iii** older than 7
 - iv** 5 or older
 - v** 8 or younger.



- 8** A die was rolled multiple times and the frequencies of outcomes counted.
- a** Add columns for relative frequency and cumulative relative frequency
 - b** How many rolls of the die were made?
 - c** Draw a cumulative relative frequency histogram and polygon.
 - d** Based on these results, find the probability of rolling:
 - i** a 6
 - ii** less than 3
 - iii** more than 4.

Outcome	Frequency
1	21
2	18
3	15
4	24
5	16
6	12

9 In a chance experiment, a coin and a die were tossed together. H = heads, T = tails.

Result	Frequency
H, 1	8
H, 2	3
H, 3	5
H, 4	7
H, 5	1
H, 6	6
T, 1	7
T, 2	12
T, 3	4
T, 4	9
T, 5	8
T, 6	6

Find the probabilities based on this experiment of getting:

- a a head and 6
- b an odd number and a tail
- c a tail and 5 or more
- d less than 4 and a head.

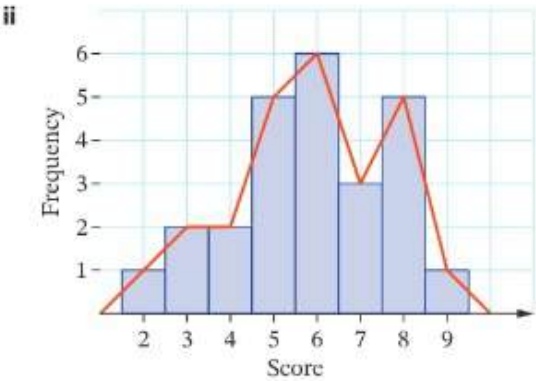
10 In an experiment where a coin was tossed 250 times, the results were 97 heads and 153 tails. Explain this result.

Exercise 12.07

- 1 a discrete b continuous c discrete
d discrete e continuous f discrete
g continuous h continuous i discrete
j discrete k discrete l discrete
- 2 a $X = \{0, 1\}$ b $X = \{0, 1, 2, \dots, 10\}$
c $X = \{0, 1, 2, 3, \dots, 20\}$ d $X = \{0, 1, 2, \dots, 31\}$
e $X = \{2, 3, 4, \dots, 12\}$

3 a i

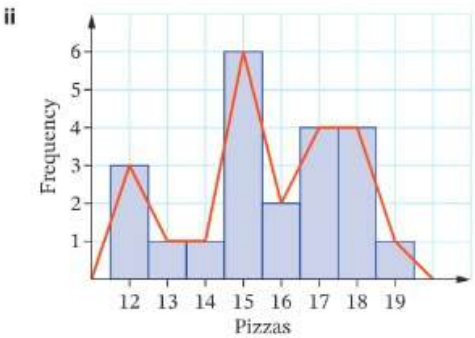
Score	Frequency
2	1
3	2
4	2
5	5
6	6
7	3
8	5
9	1



- iii highest 9, lowest 2
iv 6

b i

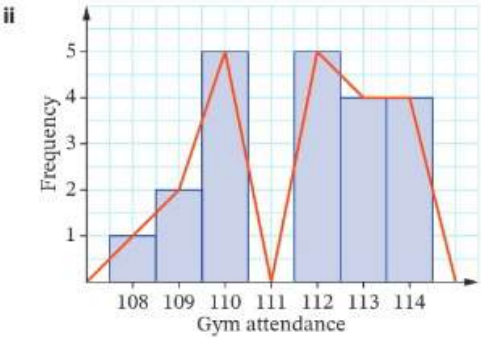
Pizzas	Frequency
12	3
13	1
14	1
15	6
16	2
17	4
18	4
19	1



- iii highest 19, lowest 12 iv 15

c i

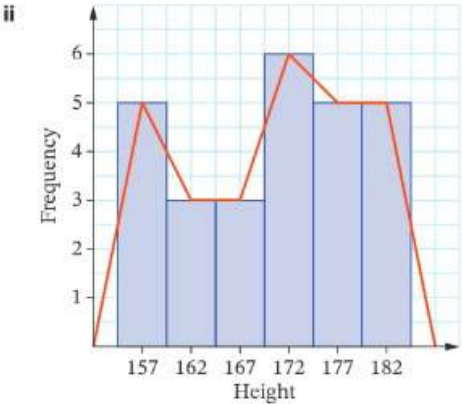
Gym attendance	Frequency
108	1
109	2
110	5
111	0
112	5
113	4
114	4



- iii highest 114, lowest 108
iv 110 and 112

d i

Height (cm)	Class centre	Frequency
155–159	157	5
160–164	162	3
165–169	167	3
170–174	172	6
175–179	177	5
180–184	182	5

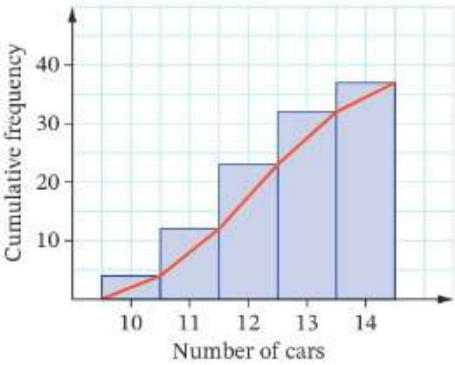


- iii highest 182, lowest 157 iv 172

4 a i

Number of cars	Frequency	Cumulative frequency
10	4	4
11	8	12
12	11	23
13	9	32
14	5	37

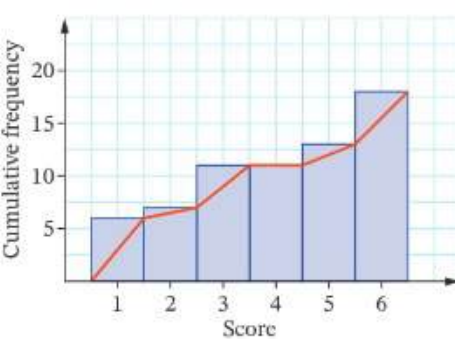
ii



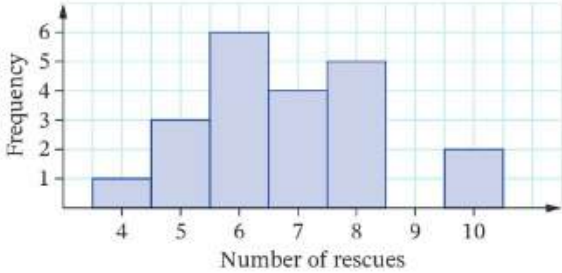
b i

Score	Frequency	Cumulative frequency
1	7	7
2	1	8
3	3	11
4	0	11
5	2	13
6	5	18

ii



5 a



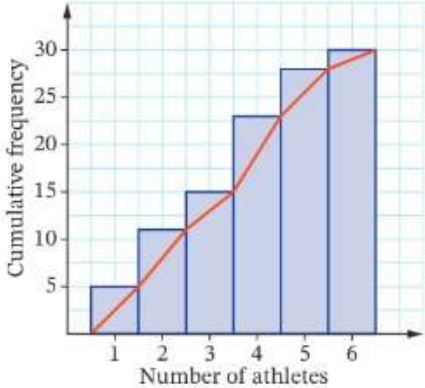
b 11 times

- 6 a i 5 ii 5 b i 4 ii 5
c i 18 ii 17 d i 4 ii 5.5
- 7 a i 6 ii 6 b i 52 ii 51
c i 18 ii 20 d i 104 ii 104
- 8 a 3 b 1.5
c 3.5 d 3
- 9 a 3 b 1 c 2 d 2

10 a i

Number of athletes	Frequency	Cumulative frequency
1	5	5
2	6	11
3	4	15
4	8	23
5	5	28
6	2	30

ii

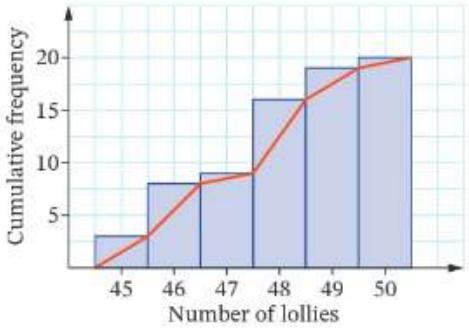


iii 3.5

b i

Number of lollies	Frequency	Cumulative frequency
45	3	3
46	5	8
47	1	9
48	7	16
49	3	19
50	1	20

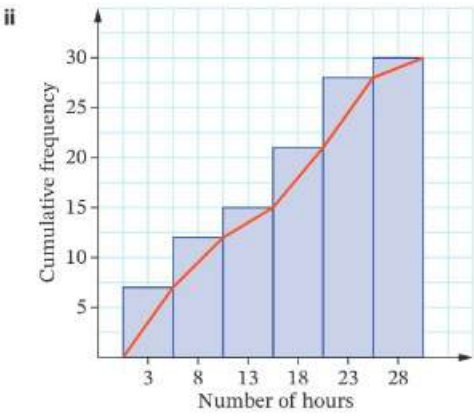
ii



iii 48

c i

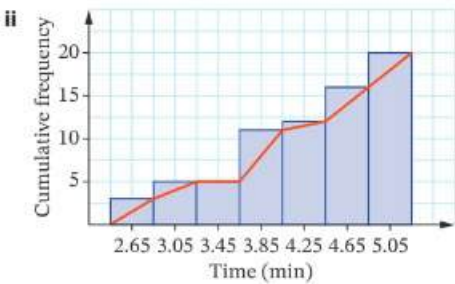
Number of hours	Class centre	Frequency	Cumulative frequency
1-5	3	7	7
6-10	8	5	12
11-15	13	3	15
16-20	18	6	21
21-25	23	7	28
26-30	28	2	30



iii 15.5

d i

Time (min)	Class centre	Frequency	Cumulative frequency
2.5–2.8	2.65	3	3
2.9–3.2	3.05	2	5
3.3–3.6	3.45	0	5
3.7–4.0	3.85	6	11
4.1–4.4	4.25	1	12
4.5–4.8	4.65	4	16
4.9–5.2	5.05	4	20

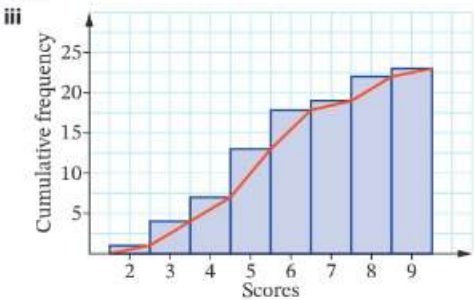


iii 4.0

11 a i

Score	Frequency	Cumulative frequency
2	1	1
3	3	4
4	3	7
5	6	13
6	5	18
7	1	19
8	3	22
9	1	23

ii 5

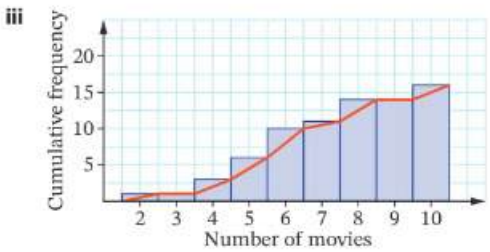


iv 5.5

b i

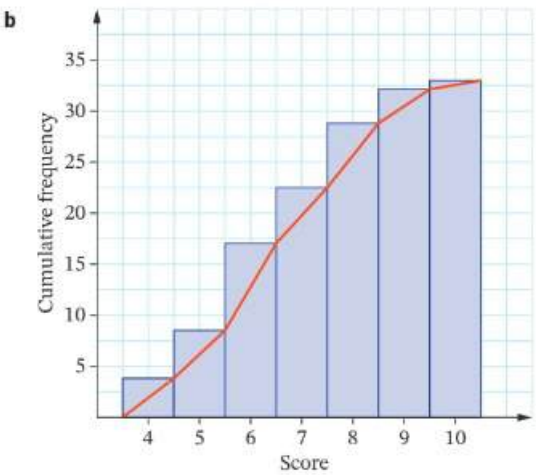
Number of movies	Frequency	Cumulative frequency
2	1	1
3	0	1
4	2	3
5	3	6
6	4	10
7	1	11
8	3	14
9	0	14
10	2	16

ii 6



iv 6

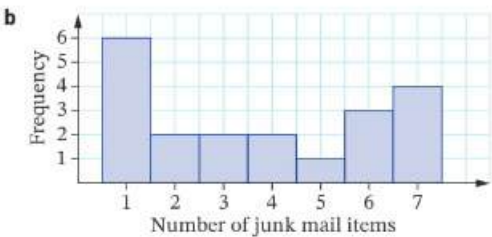
12 a 6



median = 7

13 a

Number of junk mail items	Frequency
1	6
2	2
3	2
4	2
5	1
6	3
7	4



c mode = 1, median = 3.5

Exercise 12.08

1 a

Score	Frequency	Relative frequency
4	6	$\frac{6}{23}$
5	4	$\frac{4}{23}$
6	1	$\frac{1}{23}$
7	7	$\frac{7}{23}$
8	2	$\frac{2}{23}$
9	3	$\frac{3}{23}$

b i $\frac{2}{23}$ **ii** $\frac{11}{23}$ **iii** $\frac{17}{23}$

c i 7 **ii** 6

2 a

Number of days	Frequency	Relative frequency
1	3	15%
2	6	30%
3	1	5%
4	7	35%
5	2	10%
6	1	5%

b 4

c i 15% **ii** 10% **iii** 40%

iv 50% **v** 45%

3 a

Class	Frequency	Relative frequency
0–19	9	0.18
20–39	12	0.24
40–59	18	0.36
60–79	7	0.14
80–99	4	0.08

b i 0.24 **ii** 0.22 **iii** 0.42

4 a

Velocity (m/s)	Frequency	Relative frequency
2–4	2	$\frac{1}{10}$
5–7	7	$\frac{7}{20}$
8–10	4	$\frac{1}{5}$
11–13	1	$\frac{1}{20}$
14–16	6	$\frac{3}{10}$

b i $\frac{7}{20}$ **ii** $\frac{1}{20}$ **iii** $\frac{1}{5}$

iv $\frac{7}{20}$ **v** $\frac{13}{20}$

c i $\frac{9}{20}$ **ii** $\frac{9}{10}$ **iii** $\frac{11}{20}$

5 a 10%

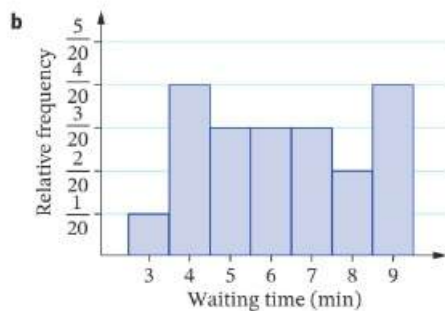
b

Sales/min	Frequency	Relative frequency
0	4	13.3%
1	12	40%
2	6	20%
3	3	10%
4	0	0%
5	5	16.7%

c 1 **d i** 20% **ii** 16.7% **iii** 26.7%

6 a

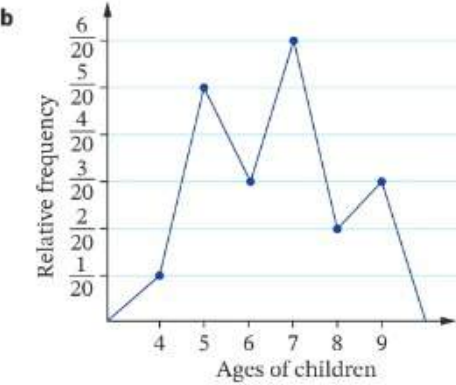
Score	Frequency	Relative frequency	Cumulative relative frequency
3	1	$\frac{1}{20}$	$\frac{1}{20}$
4	4	$\frac{1}{5}$	$\frac{1}{4}$
5	3	$\frac{3}{20}$	$\frac{2}{5}$
6	3	$\frac{3}{20}$	$\frac{11}{20}$
7	3	$\frac{3}{20}$	$\frac{7}{10}$
8	2	$\frac{1}{10}$	$\frac{4}{5}$
9	4	$\frac{1}{5}$	1



c i $\frac{3}{20}$ **ii** $\frac{3}{10}$ **iii** $\frac{1}{4}$ **iv** $\frac{7}{10}$

7 a

Score	Frequency
4	1
5	5
6	3
7	6
8	2
9	3

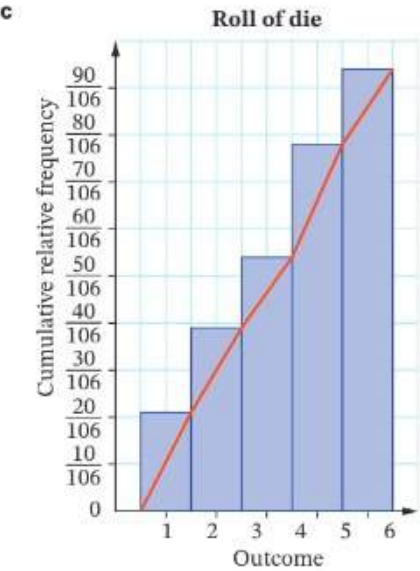


- c** **i** 0.1 **ii** 0.7 **iii** 0.25
iv 0.95 **v** 0.85

8 a

Relative frequency	Cumulative relative frequency
$\frac{21}{106}$	$\frac{21}{106}$
$\frac{9}{53}$	$\frac{39}{106}$
$\frac{15}{106}$	$\frac{27}{53}$
$\frac{12}{53}$	$\frac{39}{53}$
$\frac{8}{53}$	$\frac{47}{53}$
$\frac{6}{53}$	1

b 106



- d** **i** $\frac{6}{53}$ **ii** $\frac{39}{106}$ **iii** $\frac{14}{53}$

- 9 a** $\frac{3}{38}$ **b** $\frac{1}{4}$
c $\frac{7}{38}$ **d** $\frac{4}{19}$

10 While theoretically $P(H) = P(T) = \frac{1}{2}$, in an experiment anything can happen. Often the probabilities might come close to these in an experiment, but not always.